Over-the-Air Phase Calibration for Cooperative UAV Sensing

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***Abstract*—This paper studies over-the-air phase calibration for cooperative uav sensing. Distributed nodes exchange reciprocal pilots before coherent UAV echo combining; calibration consumes dwell and phases drift afterward. Calibration minimizes phase MSE without weighting downstream target information, while coherent sensing assumes perfect phase. The proposed mechanism is: Calibration graph, energy, and timing minimize future sensing loss under phase drift. $\phi_m(t)$ is oscillator phase, $\mathcal{G}_{cal}$ the calibration graph, and q_m drift intensity. We give a closed signal model, retain all material nuisance parameters, state the information chronology, derive the task metric, and formulate an explicit constrained design problem. After one phase gauge, the calibration graph must be connected and the target combining vector nonzero. The novelty is evaluated through matched baselines and the falsification condition: The claim fails if average phase-MSE minimization always minimizes target CRLB. Algorithms and simulations are intentionally outside this manuscript.**

***Index Terms*—integrated sensing and communication, unmanned aerial vehicle, physical layer, signal model, identifiability, resource allocation.**

I. Introduction

Low-altitude aerial networks force communication and sensing decisions onto the same waveform, aperture, and time budget. A journal-level contribution must do more than append a sensing utility to a UAV trajectory problem: it must identify the role of each node, state what is observed, retain the nuisances that remove information, and explain why the proposed design changes that observation. Broad ISAC and UAV foundations are established in [1]–[4]. Recent work has moved toward standard signals, payload reuse, distributed sensing, and explicit synchronization [5]–[7], [9]–[11], [13], [14]. The local evidence audit for this portfolio covers 434 UAV/ISAC-related PDFs through August 2026.

A. Operational Motivation

Distributed nodes exchange reciprocal pilots before coherent UAV echo combining; calibration consumes dwell and phases drift afterward. This is a problem in distributed multi-node sensing, and the raw data are local matched-filter observations. The role statement is deliberate: a UAV may be transmitter, receiver, illuminator, relay, communication user, cooperative node, or target, but those roles are never silently conflated. The wireless service remains in the model instead of being suspended for a radar-only interval unless a displayed resource variable explicitly allocates such an interval.

B. Closest Work and Exact Gap

Recent OTA synchronization estimates phases; this problem weights calibration error by future UAV sensing impact. Calibration minimizes phase MSE without weighting downstream target information, while coherent sensing assumes perfect phase. The gap is not inferred from keyword absence. A crowded topic can be crowded because it is important, in which case the novelty burden is higher. A sparse topic can instead indicate weak operational demand. This idea is classified as *crowded important*; the ranking accounts for significance, model closure, physical-layer depth, and execution risk rather than rewarding low paper count.

C. Proposed Mechanism and Contributions

Calibration graph, energy, and timing minimize future sensing loss under phase drift. The contributions are:

- A role-specific signal and observation model is stated, with $\phi_m(t)$ is oscillator phase, $\mathcal{G}_{cal}$ the calibration graph, and q_m drift intensity.
- Identifiability is checked before optimization: After one phase gauge, the calibration graph must be connected and the target combining vector nonzero.
- The primary problem uses the explicit variable set $\mathcal{D} = \{\mathcal{G}_{cal}, p_{mn}, t_c\}$ and a matched-resource falsification condition: The claim fails if average phase-MSE minimization always minimizes target CRLB.

The manuscript stops after introduction, system model, and problem formulation. It makes no algorithmic or numerical performance claim.

II. System and Signal Model

A. Geometry, Roles, and Time-Frequency Organization

All node reference points are expressed in one right-handed Cartesian frame. Positions and velocities are constant only over the interval in which that assumption is stated. Large-scale gains are deterministic functions of geometry within a frame. Small-scale gains and nuisances follow the displayed likelihood or a declared bounded set. Time, subcarrier, antenna, link, and state indices are retained whenever removing them would change observability.

$$\mathbf{x}_m = \mathbf{W}_m \mathbf{s}_m, \qquad \sum_m \mathrm{tr}(\mathbf{W}_m \mathbf{W}_m^H) \le P_{\mathrm{net}}. \tag{1}$$

Equation (1) defines the transmitted waveform, state primitive, or causal information evolution. Communication symbols are normalized to unit average energy before precoding. Total energy is measured after precoding. Scheduling, pilot, sensing, calibration, and feedback samples consume the same frame budget and are never counted twice.

B. Received Signal and Nuisance Structure

$$\mathbf{z}_\ell = \mathbf{h}_\ell(\boldsymbol{\xi};\mathcal{D}) + \mathbf{B}_\ell\boldsymbol{\eta} + \mathbf{v}_\ell, \qquad \mathbf{v}_\ell \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_\ell(\mathcal{D})), \tag{2}$$

The desired parameter is denoted by $\boldsymbol{\xi}$ or by the problem-specific state in $\phi_m(t)$ is oscillator phase, $\mathcal{G}_{cal}$ the calibration graph, and q_m drift intensity. Direct channels, complex reflectivities, common phases, clock offsets, payload symbols, class amplitudes, and hardware states are nuisances unless explicitly part of the desired output. Gaussian receiver noise is proper with positive-definite covariance. A quantized, finite-state, or mixture likelihood replaces that Gaussian law wherever the physical front end requires it.

C. Problem-Specific Physical Law

The generic observation above is specialized by the following defining relation:

$$\mathbf{C}_\phi = (\mathbf{L}_{cal}(\mathbf{p}) + \mathbf{Q}_{drift}^{-1})^\dagger, \quad G_{coh} = \mathbb{E}|\sum_m g_m e^{j\widetilde{\phi}_m}|^2 \tag{3}$$

Equation (3) is the mathematical source of novelty. It is not an extra penalty attached to a generic model. It changes either the likelihood, equivalent information, ambiguity set, state recursion, or causal feasible set. Every symbol in it is defined in the role statement, unknown list, or the paragraph immediately surrounding the equation.

D. Communication and Sensing Metrics

For a scheduled user u, the communication requirement is evaluated as $R_u = B_u \log_2(1+\gamma_u)$ for a long coded block, or by the displayed finite-blocklength/queue expression when that approximation is invalid. Interference produced by the sensing component is included in γ_u unless the receiver is explicitly assumed to know and cancel it.

Detection designs fix false alarm before comparing miss probability, noncentrality, or divergence. Estimation/localization designs partition the FIM as

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_{\xi\xi} & \mathbf{J}_{\xi\eta} \\ \mathbf{J}_{\eta\xi} & \mathbf{J}_{\eta\eta} \end{bmatrix}, \qquad \mathbf{J}_\xi^e = \mathbf{J}_{\xi\xi} - \mathbf{J}_{\xi\eta}\mathbf{J}_{\eta\eta}^\dagger\mathbf{J}_{\eta\xi}, \tag{4}$$

only after fixing all gauges. Tracking designs propagate process uncertainty before adding action-dependent information. Thus nominal received SNR is never reported as a substitute for the actual task metric.

E. Validity Domain and Limiting Cases

Zero drift favors early calibration; high drift favors late calibration; equal node gains approach average phase MSE. The model deliberately excludes unconstrained real clutter, proprietary commercial hardware responses, and empirical scene fitting. Such effects may be used in later stress testing but cannot create the novelty. The claimed domain is the parametric waveform, channel, motion, clock, fading, or hardware model stated above.

III. Problem Formulation

A. Decision Variables and Primary Problem

The complete decision set is $\mathcal{D} = \{\mathcal{G}_{cal}, p_{mn}, t_c\}$. Variables are selected using only information available before the corresponding observation. The primary problem is

$$\begin{aligned} \underset{\mathcal{D}=\{\mathcal{G}_{cal},p_{mn},t_c\}}{\text{minimize}} \quad & -\min_{\boldsymbol{\xi}\in\Xi} \mathbb{E}|\sum_m g_m(\boldsymbol{\xi}) e^{j\widetilde{\phi}_m(t_s)}|^2 && \text{(5a)} \\ \text{subject to} \quad & \mathcal{G}_{cal} \text{ connected} && \text{(5b)} \\ & \sum_{(m,n)} p_{mn}T_{mn} \le E_{cal} && \text{(5c)} \\ & T_{cal} + T_{sense} \le T_{frame} && \text{(5d)} \\ & R_u \ge \bar{R}_u && \text{(5e)} \end{aligned}$$

The objective is stated in the units of detection information, estimation error, tracking covariance, latency, energy, rate, or a dimensionally declared weighted combination. No unbounded multiplier can replace a hard service, rank, or hardware requirement.

B. Constraint-by-Constraint Interpretation

1) Constraint 1, $\mathcal{G}_{cal}$ connected. This is a service or reliability requirement evaluated with the same resources as sensing.
2) Constraint 2, $\sum_{(m,n)} p_{mn}T_{mn} \le E_{cal}$. This is a physical resource or standards constraint and cannot be relaxed by normalization.
3) Constraint 3, $T_{cal} + T_{sense} \le T_{frame}$. This restriction makes the hardware, role, timing, or motion decision implementable.
4) Constraint 4, $R_u \ge \bar{R}_u$. This condition protects identifiability or excludes a formally attractive but information-singular design.

Together these constraints prevent a trivial sensing-only solution, unlimited power, noncausal side information, impossible simultaneous node roles, or a numerically regularized but unidentifiable design.

C. Identifiability, Feasibility, and Gauge Removal

After one phase gauge, the calibration graph must be connected and the target combining vector nonzero. Resource feasibility and information feasibility are different. Power, time, bandwidth, fronthaul, and motion budgets may all be satisfied while the observation Jacobian remains rank deficient or the two hypothesis families remain identical. Conversely, a full-rank model can be operationally infeasible under latency or communication QoS. The formulation reports both failures rather than hiding either behind an optimizer.

D. Required Analytical Results and Matched Baselines

Derive sensing-weighted effective resistance, symmetric optimal timing, and coherence break-even. A complete follow-on article should establish these results before a large-scale numerical algorithm. The mandatory baselines are: Uniform complete graph, minimum phase-MSE calibration, and non-coherent fusion. They must use identical energy, bandwidth, antennas, coherent time, communication demand, synchronization, and side information. The claim fails if average phase-MSE minimization always minimizes target CRLB.

E. Information Chronology and Admissible Policies

All geometry priors, calibrated uncertainty sets, and communication demands are declared before the frame. A design action may use past received samples, decoder outputs, queue states, and state estimates, but not the target realization or decoder decision produced by the action itself. Known pilots, soft payload posteriors, hard-decoded payloads, and unavailable random data are therefore different objects in the likelihood. Node clocks, carrier phases, reflectivities, direct leakage, and platform states are estimated, integrated out, or eliminated as stated; none is inserted at its true value.

F. Model Closure and Dimensional Consistency

The observation vector is formed by stacking only the time, frequency, antenna, and link samples selected by $\mathcal{D}$. Its covariance is positive definite on the retained subspace. Complex gains are represented by real and imaginary parts before a Fisher matrix is formed. A gauge is removed before a pseudoinverse is used. The dimensions of every Jacobian therefore follow from the declared number of desired and nuisance parameters. A singular equivalent information matrix is reported as nonidentifiability rather than regularized into a finite bound.

G. Boundary Regimes and Falsification Protocol

Three limits must be checked. The communication-only limit must reproduce the accepted wireless model. The sensing-only or perfectly known-nuisance limit is an optimistic upper bound, not the proposed operating point. The lowest-resource limit must become infeasible when a rank, reliability, or latency condition is violated. Baselines are matched in radiated energy, bandwidth, antennas, coherent duration, communication load, synchronization, and side information. If the proposed gain disappears under that matching, the mechanism is falsified and should not be rescued by adding unrelated technologies.

H. Required Path From Formulation to a Full Paper

A follow-on study should first prove the stated identifiability and scaling results, then derive a computable bound and at least one structural solution in a reduced regime. Only afterward should a nonconvex algorithm be introduced. Numerical work should vary the physical parameter responsible for novelty, include the infeasible boundary, and compare against all stated matched baselines. No empirical clutter trace, proprietary hardware response, or open-ended learning model is needed to establish the core claim.

I. Parameter Partition and Local Identifiability Audit

Let $\boldsymbol{\vartheta} = [\boldsymbol{\xi}^T, \boldsymbol{\eta}^T]^T$ collect the desired state and all nuisance quantities. The desired block is fixed by the title and task; the nuisance block contains every unknown complex gain, direct-link coefficient, clock or carrier offset, payload uncertainty, hardware state, or process parameter that appears in the observation. The score vector is formed from the actual likelihood,

$$\mathbf{s}_{\vartheta}(\mathbf{y}) = \partial_{\boldsymbol{\vartheta}} \log p(\mathbf{y}|\boldsymbol{\vartheta}, \mathcal{D}), \qquad \mathbf{J}_{\vartheta} = \mathbb{E}[\mathbf{s}_{\vartheta}\mathbf{s}_{\vartheta}^T]. \tag{6}$$

For deterministic bounded nuisances, the same partition is applied to the local Jacobian or to the worst-case likelihood family. The rank test is carried out before inversion and at representative interior points of the declared state region, not only at one favorable nominal geometry. This audit directly implements the idea-specific condition stated above. It also reveals whether an apparent performance gain comes from information about the desired state or from treating a nuisance as known.

Gauge freedoms are handled explicitly. One network clock, common carrier phase, global translation, or global rotation is fixed only when the observation is invariant to that transformation. Removing a gauge does not create physical information; it merely selects coordinates on an equivalence class. If the remaining desired block is singular, the correct conclusion is that the proposed node roles, samples, or geometry are insufficient.

J. Unified Resource Ledger

All comparisons use one resource vector

$$\mathbf{r}(\mathcal{D}) = [E_{\mathrm{rad}}, T_{\mathrm{air}}, B_{\mathrm{occ}}, N_{\mathrm{RF}}, C_{\mathrm{FH}}, E_{\mathrm{prop}}, I_{\mathrm{side}}]^T, \tag{7}$$

where the entries denote radiated energy, occupied airtime and bandwidth, active RF chains, coordination/fronthaul load, propulsion energy, and side information. Components irrelevant to a particular idea are set to zero rather than silently omitted. Pilot, calibration, sensing, feedback, ACK, and relay slots are counted in T_{air}; an observation reused by communication and sensing is counted once. Decoder posteriors, synchronized clocks, known geometry, and perfect direct-path references are represented in I_{side} so that a baseline cannot receive stronger side information for free.

The communication metric and sensing metric are therefore evaluated at the same $\mathbf{r}(\mathcal{D})$. A radar-only covariance with no data symbols may be reported as an upper bound, but not as the principal comparison. Likewise, a communication-only design may establish the opportunity cost but cannot satisfy an information-rank constraint by definition.

K. Research Hypotheses

The formulation supports three testable hypotheses. First, the mechanism-level claim is that Calibration graph, energy, and timing minimize future sensing loss under phase drift. Second, the information claim is that After one phase gauge, the calibration graph must be connected and the target combining vector nonzero. Third, the operational claim is that the improvement survives communication QoS and the complete resource ledger. The predeclared falsification test is: The claim fails if average phase-MSE minimization always minimizes target CRLB.

These hypotheses are intentionally narrower than a generic assertion that joint design outperforms separate design. The first should be tested by varying the physical parameter that activates the mechanism. The second should be tested at and across the rank boundary. The third should be tested by matching all entries of (7). A result that improves only because of extra pilots, a longer CPI, perfect synchronization, or stronger side information does not support the proposed contribution.

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